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Making a playable simulation of markets and organisations

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1 Motivation

During Economy Week (Deutsch: Wirtschaftwoche), as an exercise in *business administration*, we *played* a game, where you were the board of a company manufacturing some arbitrary product and had to make decisions to increase sales, lower cost and generally beat out your competition. For me, the aspect of being able to come up with novel management ideas and compete with your friends really hooked me into the concept. This may sound hyperbolic, but during that week I genuinely woke up excited to go to school, not to actually learn stuff (and whatever) but just to play a few more rounds and (aspirationally) completely destroy my competition. The game we got to play during Economy Week, in my opinion, had a tantalising hook but the interface was utterly hostile to non-directed interaction: it was pretty much an excel table! Anyway, I set off to make a better, deeper version of this simulation.

"We do this not because it is easy, but because we thought it would be easy"

Not JFK.

This is the running motto of this project.

2 Goals

The main goals of this project are to make

1. A multiplayer game where you lead a company within a market
2. Have it work
3. Document what I learnt, how and why I did what I did

3 Concept

The game itself is played in 2 repeating phases:

1. the decisions phase,
2. the simulation phase

3.1 Decisions phase

During the decisions phase you act as the board of the company, with absolute control over your company.

The players all start out with a shell of a company and a pittance of money. You then design your product (e.g.: a coffee machine), choosing parts and determining the properties of said product, ideally, carving out a niche for your company. The players must make sure to effectively market your product so that the population actually know that it exists and can thus buy it.

In later stages of the game the players are finally able to afford high tech production equipment and manufacturing techniques. This way can develop their products and market them so that they cover large sections of the market more effectively and efficiently. For example by purchasing machines that produce products quicker, which, however, are also far more expensive than the standard ones. Companies can also choose to offer multiple products at once allowing them to cover multiple segments of the market.

In the end players lose when either

1. Their company goes bankrupt.
2. Another player gets a controlling share in their company: at which point the companies are merged (under the control of the bigger company), causing the smaller company's former owner to lose.

3.2 Simulation phase

After each company has submitted their decisions ending the decisions phase, the economic simulation runs, calculating whether employees at the company are satisfied enough to actually stay at the company, computing the production of products, running of the company (based on player decisions) and, finally, simulating the marketing and purchasing of products, determining whether the company was successful in its job.

4 Technology

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5 Iteration

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6 Economic model

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7 Actually making it fun

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